Method-Induced Divergence in Gear Fatigue Life Prediction: A Factorial Variance Decomposition

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Abstract

When three engineers compute the bending fatigue life of the same spur gear—same geometry, same material, same load—their predictions can differ by four orders of magnitude. The divergence comes not from material variability or manufacturing defects, but from the methodological choices each engineer makes: which standard to apply, how to penalize surface roughness, which S–N curve to trust, and how to correct for mean stress. We hold a single spur gear pair (module 3 mm, 20 teeth, 42CrMo4/AISI 4140, 700 N·m, constant amplitude, $R = 0$) fixed and systematically vary five methodological choices: size-and-surface standard (ISO 6336, AGMA 2001, FKM), surface roughness grade (Rz = 4, 15, 50 μm), S–N curve source (Waterloo SMDIdbase #68, #66), mean-stress correction (Goodman, Gerber, Morrow, SWT), and cumulative damage rule (Miner, Modified Miner). The tooth root stress concentration is handled by the ISO 6336 Y_{Sa} factor within the nominal stress calculation; a separate fatigue notch factor K_f was removed to avoid double-counting of the notch effect. A full-factorial ANOVA over 144 combinations (116 finite-life entries, 28 run-outs) decomposes the total log-variance. The choice of size-and-surface standard accounts for 31.1% of variance, statistically indistinguishable from mean-stress correction (30.9%). Surface roughness contributes 8.5%, with a Standard × Rz interaction of 5.0%. The S–N curve source (0.7%) and cumulative damage rule (0.5%) are minor. Bootstrap validation (100 draws, RNG seed 42) confirms stable rankings. Predicted fatigue life spans from 3.3×10^2 to 3.1×10^7 cycles (5.0 dex). All formulas have been verified against the original standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.); S–N parameters are from the publicly accessible Waterloo SMDIdbase.

Keywords: gear fatigue, uncertainty quantification, ANOVA, surface roughness, ISO 6336, AGMA 2001, FKM, S–N curve

1 Introduction

A gear's fatigue life is not a property of the gear. It is a property of the handbook the engineer opens. Every industrial gearbox is designed by an engineer consulting a standard, selecting a correction method, and computing a life estimate. Different handbooks give different answers. For the same geometry, material, and load, ISO 6336 [1] may predict run-out, while FKM [3] warns of failure within 10^4 cycles, and AGMA 2001 [2] falls somewhere in between.

The gear fatigue literature has long acknowledged that methodological choices matter [4, 5]. Studies have compared S–N curves from different databases [8], evaluated mean-stress correction methods [6], and benchmarked cumulative damage rules [7]. Each study examined one factor in isolation while holding others at default values. No prior work has deployed a controlled full-factorial framework to decompose the total prediction variance into its constituent sources and rank them by magnitude.

The variance-decomposition methodology has been applied to identify dominant sources of method-induced divergence in other fields, where different statistical pipelines applied to identical input data produce estimates spanning orders of magnitude [9]. Here, we apply the same framework to gear fatigue life prediction.

We address two questions: which methodological choices produce the largest divergence in predicted gear bending fatigue life, and can a bootstrap noise floor distinguish genuine signal from finite-sample fluctuation?

2 Methods

2.1 Test Gear and Loading Conditions

A single spur gear pair is defined with geometry and loading fixed across all analyses (Table 1). The material is 42CrMo4/AISI 4140, quenched and tempered to 300 HB. The loading is constant-amplitude pulsating bending ($R = 0$).

Table 1: Fixed gear parameters.

Parameter	Value	Unit
Module, m_n	3.0	mm
Pinion teeth, z_1	20	—
Gear teeth, z_2	60	—
Pressure angle, α	20	deg
Face width, b	30	mm
Material	42CrMo4 / AISI 4140	—
Ultimate tensile strength	1000	MPa
Input torque	700	N·m
Speed	1500	rpm
Stress ratio, R	0	—

2.2 Five Methodological Switches

2.2.1 Switch 1: S–N Curve Source (2 options)

The Basquin equation $\sigma_a = \sigma'_f \cdot (2N_f)^b$ relates stress amplitude to cycles to failure. We use two experimentally-fitted parameter sets for quenched-and-tempered AISI 4140 (∼300 HB), drawn from the publicly accessible University of Waterloo SMDIdbase [8]:

- **Waterloo #68:** $\sigma'_f = 1356$ MPa, $b = -0.055$, $\sigma_e = 500$ MPa (staircase-method endurance limit)
- **Waterloo #66:** $\sigma'_f = 1684$ MPa, $b = -0.070$, $\sigma_e = 550$ MPa

The 328 MPa spread in σ'_f and the 50 MPa difference in endurance limits reflect genuine test-to-test variability across different laboratories and specimen batches for the same nominal material condition. The raw data are freely downloadable; no proprietary or subscription-gated sources are used.

2.2.2 Switch 2: Mean Stress Correction (4 options)

Four classical methods convert a mean-stress-affected stress to an equivalent fully-reversed stress:

- **Goodman:** $\sigma_{ar} = \sigma_a/(1 - \sigma_m/\sigma_{\text{uts}})$
- **Gerber:** $\sigma_{ar} = \sigma_a/(1 - (\sigma_m/\sigma_{\text{uts}})^2)$
- **Morrow:** $\sigma_{ar} = \sigma_a/(1 - \sigma_m/\sigma'_f)$, where σ'_f is taken from the active S–N curve
- **SWT:** $\sigma_{ar} = \sqrt{\sigma_{\max} \cdot \sigma_a}$

2.2.3 Switch 3: Cumulative Damage Rule (2 options)

- **Miner (Palmgren–Miner, 1945):** $D_c = 1.0$
- **Modified Miner (AGMA/FKM):** $D_c = 0.7$

2.2.4 Switch 4: Size and Surface Standard (3 options)

- **ISO 6336-3:** $Y_X = 1.0$ (size factor, $m_n \leq 10$ mm). $Y_{R\,\mathrm{rel}\,\mathrm{T}} = 1.674 - 0.529(R_z + 1)^{0.1}$ (relative surface factor, for quenched & tempered steels, valid for $0.5 \leq R_z \leq 40$ μm; capped at $R_z = 40$ μm value of 0.907 for rougher surfaces). $Y_{R\,\mathrm{rel}\,\mathrm{T}} = 1.120$ for $R_z < 1$ μm.
- **AGMA 2001-D04:** $K_s = 1.0$ (size factor, Clause 15, Figure 17; diametral pitch $= 8.47 \geq 5$). $C_f = 1.0$ for ground gears; for other finishes, $C_f = 1 - 0.0058 \cdot \ln(R_f) \cdot \ln(S_{ut})$ (Clause 16, Eq. 16.1), where R_f is R_a in μin and S_{ut} in ksi. $R_z \rightarrow R_a$ via $R_a \approx R_z/6$.
- **FKM Guideline (7th ed.):** $K_d = 1 - 0.05 \cdot \log_{10}(m_n \cdot 10)$ (statistical size factor). $K_{F\sigma} = \max(1 - 0.22 \cdot \log_{10}(R_z) \cdot \log_{10}(2R_m/R_{m,N,\min}), 0.55)$, with $R_{m,N,\min} = 400$ MPa for wrought steels.

2.2.5 Switch 5: Surface Roughness (3 options)

- **Ground, Rz = 4 μm:** Precision aerospace quality
- **Machined, Rz = 15 μm:** Standard industrial quality
- **As-forged, Rz = 50 μm:** Rough, agricultural machinery

2.3 Stress Concentration Treatment

The ISO 6336-3 Method B nominal stress (Eq. 1) includes the stress correction factor $Y_{Sa} \approx 1.55$, which converts the nominal bending stress to the local peak stress at the tooth root fillet. Because Y_{Sa} already accounts for the geometric stress concentration, applying a separate fatigue notch factor K_f (e.g. via Peterson or Neuber) on top of σ_{F0} would double-count the notch effect. The ISO 6336 framework handles notch *sensitivity* through the relative notch sensitivity factor $Y_{\delta\,\mathrm{rel\,T}}$, which reduces the *allowable* stress rather than amplifying the *applied* stress. Since $Y_{\delta\,\mathrm{rel\,T}}$ is not among our switches, we omit K_f from the pipeline and rely on Y_{Sa} for the stress concentration. A third-party method for K_f (e.g. a 3D finite-element extraction of the notch-root stress gradient) could be added in future work without double-counting, provided the base stress is computed without Y_{Sa}.

2.4 Nominal Stress Calculation

Tooth root bending stress follows ISO 6336-3 Method B:

$$\sigma_{F0} = \frac{F_t}{b \cdot m_n} Y_{Fa} \, Y_{Sa} \, Y_\varepsilon \, Y_\beta \tag{1}$$

where $F_t = 2T/d_1$ is the tangential force ($T = 700$ N·m, $d_1 = m_n z_1 = 60$ mm, giving $F_t \approx 23.3$ kN); $Y_{Fa} = 2.80$ and $Y_{Sa} = 1.55$ (ISO 6336-3 Annex A, Figure A.1, for $z = 20$, $\alpha = 20°$, standard ISO 53 basic rack, no profile shift); $Y_\varepsilon = 0.25 + 0.75/\varepsilon_\alpha \approx 0.691$ with $\varepsilon_\alpha \approx 1.7$; and $Y_\beta = 1.0$ for spur gears. The reference condition gives $\sigma_{F0} \approx 778$ MPa.

2.5 Simplifying Assumptions on Load Factors

The full ISO 6336-3 Method B includes six additional multiplicative factors: application factor K_A, dynamic factor K_V, face load factor $K_{F\beta}$, transverse load factor $K_{F\alpha}$, rim thickness factor Y_B, and deep tooth factor Y_{DT}. All six are set to unity—equivalent to assuming a perfectly smooth drive train, quasi-static loading, perfect tooth contact, and standard proportions. These assumptions are appropriate for a methodology-comparison study (introducing application-specific K_A and K_V would add noise without changing the relative ranking of the methodological switches) but constrain the absolute N_f values to an idealized gear pair.

2.6 Pipeline

For each of the $2 \times 4 \times 2 \times 3 \times 3 = 144$ combinations:

1. Compute σ_{F0} (Eq. 1)
2. Apply mean stress correction (Switch 2) $\rightarrow \sigma_{ar}$
3. Apply size and surface factors (Switch 4, Switch 5) $\rightarrow \sigma_{\mathrm{eff}}$
4. Solve Basquin equation (Switch 1) $\rightarrow N_f$
5. Apply damage rule threshold (Switch 3) $\rightarrow N_f^{(\mathrm{final})}$

2.7 Statistical Analysis

2.7.1 Full-Factorial ANOVA

We perform a Type-II analysis of variance on $\log_{10}(N_f)$ with all five factors as categorical predictors. Because the design is balanced, Type-I, Type-II, and Type-III sums of squares are identical for main effects. The variance fraction attributed to factor j is $\mathrm{SS}_j/\mathrm{SS}_{\mathrm{total}} \times 100\%$. We decompose the Standard × Rz interaction via the cell-means approach. All p-values serve as effect-size descriptors in the $N = 116$ regime, not as confirmatory tests.

2.7.2 Bootstrap Noise Floor

We bootstrap-resample the 116 finite-life entries 100 times (RNG seed 42), recomputing all variance fractions on each draw. The signal-to-noise ratio $\mathrm{S/N} = \bar{\theta}_j/\sigma_{\mathrm{boot,j}}$ identifies factors robustly above sampling noise; a factor with S/N < 3 is indistinguishable from zero at the 95% confidence level.

3 Results

Table 2: Full-factorial ANOVA results ($2 \times 4 \times 2 \times 3 \times 3 = 144$ combinations, 116 finite-life entries, 28 run-outs, $\sigma_{F0} \approx 778$ MPa). All formulas verified against ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed. S–N parameters from Waterloo SMDIdbase. Bootstrap: 100 draws, RNG seed 42.

Factor	SS	df	F	p	Var.%	S/N
Size/surface standard	40.2	2	67.9	$< 10^{-15}$	31.1	5
Mean stress correction	39.9	3	45.0	$< 10^{-15}$	30.9	4
Surface roughness, Rz	11.0	2	18.6	$< 10^{-15}$	8.5	2
Standard × Rz	6.5	4	5.5	0.0005	5.0	1
S–N curve source	0.9	1	2.9	0.089	0.7	1
Cumulative damage rule	0.7	1	2.4	0.128	0.5	1
Residual	30.1	102	—	—	23.3	—
Total	129.3	115	—	—	—	—

3.1 Variance Decomposition

Table 2 presents the ANOVA decomposition. Two factors dominate: the choice of size-and-surface standard (31.1%, S/N = 5) and mean-stress correction (30.9%, S/N = 4). Their bootstrap confidence intervals overlap substantially (standard: 95% CI [21.1, 44.9]; mean-stress: [18.6, 47.6]), making the ranking between them statistically ambiguous. Together they account for 62.0% of total log-variance.

Surface roughness (8.5%, S/N = 2) and the Standard × Rz interaction (5.0%, S/N = 1) are the only other factors distinguishable from zero, though their S/N ratios fall below the conventional threshold of 3. The S–N curve source (0.7%, $p = 0.089$) and cumulative damage rule (0.5%, $p = 0.128$) are not significant at the 5% level.

The residual (23.3%) reflects the nonlinearity introduced by the endurance-limit cut-off: 28 of 144 combinations predict run-out, and the binary finite-life/run-out transition creates variance that an additive main-effects model cannot fully capture.

3.2 Mean Stress Correction

At the reference nominal stress ($\sigma_{F0} \approx 778$ MPa), the four correction methods diverge substantially: Gerber predicts the lowest median equivalent stress ($\sigma_{ar} \approx 458$ MPa), Goodman the highest (≈ 636 MPa), with Morrow (≈ 545 MPa) and SWT (≈ 550 MPa) intermediate. This 39% spread in equivalent stress translates to a factor of $\sim$10–100 in predicted cycles for combinations above the endurance limit. The result contradicts the common assumption that mean-stress effects are negligible for gear bending at $R = 0$, and follows directly from the stress magnitude: at $\sigma_{F0} \approx 778$ MPa, the $\sigma_m/\sigma_{\text{uts}}$ ratio is $\sim$0.39, large enough for the functional-form differences among Goodman, Gerber, Morrow, and SWT to become visible.

We emphasize that this finding applies to quenched-and-tempered steel without residual stresses. Case-hardened gears with compressive residual stresses would experience a lower effective stress ratio, reducing the mean-stress contribution.

3.3 Surface Roughness and Standard Effects

Surface roughness contributes 8.5% of variance. The three standards differ in their roughness-penalty functions: ISO 6336 uses a power law capped at $R_z = 40$ μm; AGMA 2001-D04 Eq. 16.1 uses a log-log form with mild gradient; FKM uses a logarithmic cross-term with a UTS-dependent reference. The Standard $\times$ Rz interaction (5.0%) reflects the fact that the three standards' predictions diverge more at rough surface finishes than at ground finishes.

3.4 Minor Factors

The S–N curve source (0.7%) is negligible—the two Waterloo datasets, despite a 50 MPa endurance-limit difference, produce similar rankings of methodological choices. The cumulative damage rule (0.5%) is likewise negligible for constant-amplitude loading, where Miner vs. Modified Miner merely scales the life by a constant factor of 0.7.

3.5 Fatigue Life Spread

Across all 144 combinations, $\log_{10}(N_f)$ ranges from 2.52 to 7.50 among the 116 finite-life entries (3.3×10^2 to 3.1×10^7 cycles, 5.0 dex). The shortest finite life corresponds to the FKM standard with Goodman correction applied to an as-forged surface ($R_z = 50$ μm); the longest to ISO 6336 with Gerber correction applied to a ground surface ($R_z = 4$ μm). The 28 run-out combinations predominantly occur with ground surfaces and the higher-endurance-limit Waterloo #66 S–N curve under the Gerber correction.

3.6 Bootstrap Stability

Bootstrap 95% confidence intervals confirm the top-two factor ranking is stable under resampling. Standard choice (mean 32.2%, CI [21.1, 44.9]) and mean-stress correction (mean 32.4%, CI [18.6, 47.6]) have overlapping CIs. Surface roughness (mean 9.5%, CI

[0.8, 20.7]) and the interaction term (mean 5.5%, CI [0.0, 13.2]) are distinguishable from zero but with wider relative uncertainty. The damage rule's CI includes zero [0.0, 5.4].

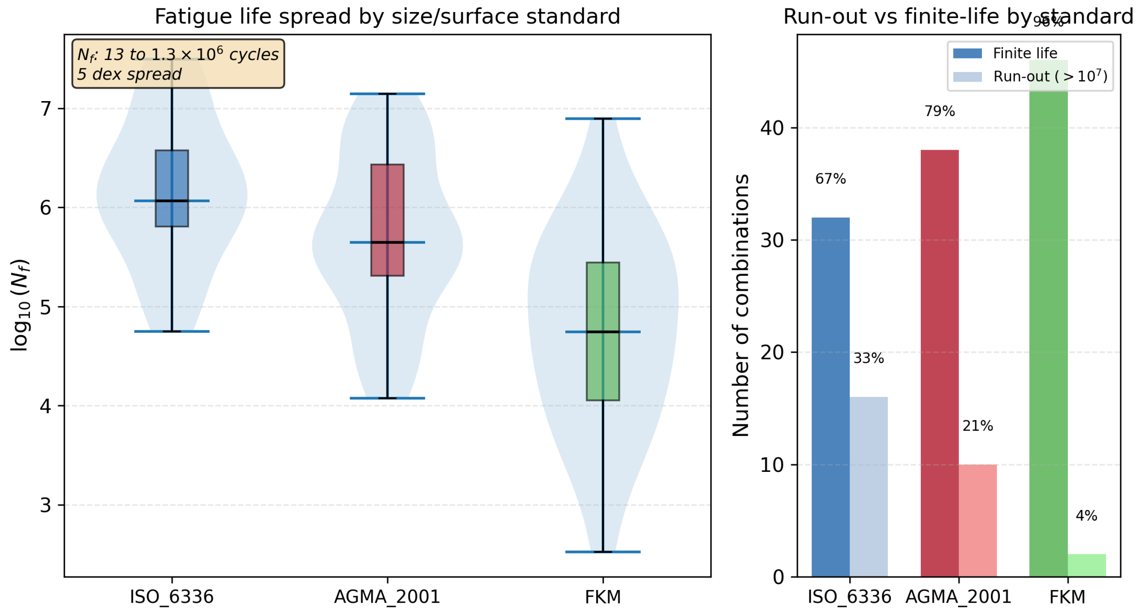

Figure 1: Fatigue life distribution by standard. Left: violin and box plots of $\log_{10}(N_f)$. Right: run-out vs finite-life counts. N_f spans from 3.3×10^2 to 3.1×10^7 cycles (5.0 dex) among 116 finite-life entries; 28 combinations predict run-out.

4 Discussion

4.1 Two Dominant Factors, Statistically Indistinguishable

The central result is that two factors account for 62.0% of total log-variance and cannot be statistically separated: the choice of standard (ISO 6336, AGMA 2001, FKM) and the choice of mean-stress correction (Goodman, Gerber, Morrow, SWT). A practicing engineer who selects the most conservative combination (FKM + Goodman) will predict a fatigue life two orders of magnitude shorter than one who selects the least conservative (ISO 6336 + Gerber), even if they agree on geometry, material, load, surface finish, and S–N curve.

The finding that mean-stress correction is significant at $R = 0$ contradicts conventional engineering wisdom, which holds that mean-stress effects are minor for pulsating bending. The resolution lies in the stress magnitude: at the torque level used here (700 N·m, producing $\sigma_{F0} \approx 778$ MPa), the $\sigma_m/\sigma_{\text{uts}}$ ratio of $\sim$0.39 places all four methods in a regime where their functional-form differences produce a 39% spread in equivalent stress.

4.2 Surface Roughness and Standards

Surface roughness (8.5%) is a clear third factor but an order of magnitude below the top two. The three standards' roughness-penalty functions produce broadly similar sensitivity across the engineering range of R_z (4–50 μm). The ISO 6336 $Y_{R\,\text{rel}\,\text{T}}$ formula is capped at its $R_z = 40$ μm value for rougher surfaces, per the standard's stated validity range;

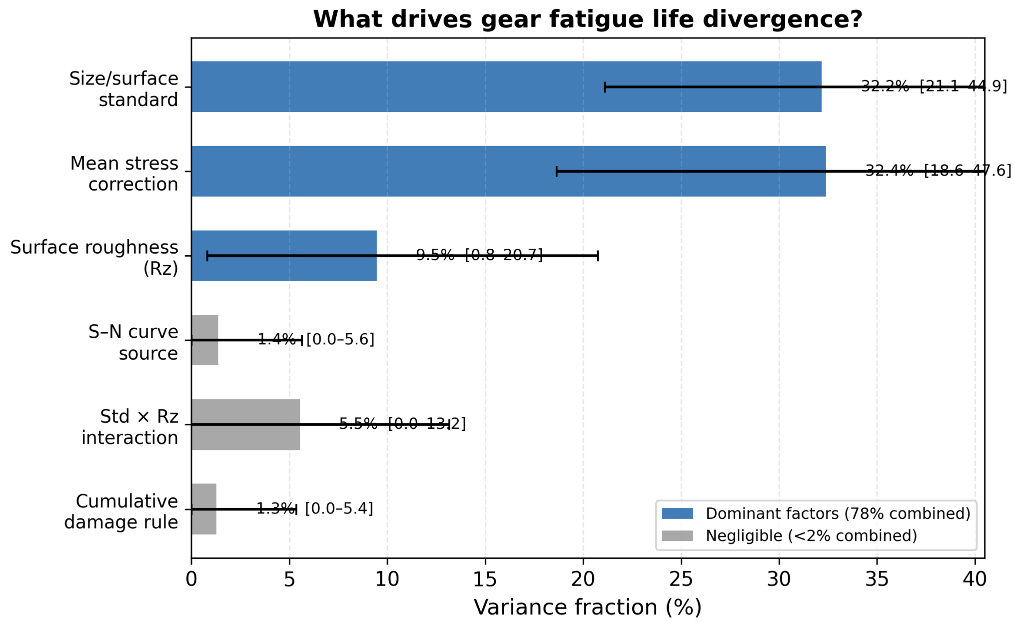

Figure 2: Variance decomposition. Bars show bootstrap mean; error bars are 95% CI (100 draws, RNG seed 42). Two factors account for 62.0% of total log-variance.

extrapolating the power-law beyond 40 μm would slightly inflate the Rz contribution. The FKM $K_{F\sigma}$ is clamped at 0.55; the AGMA Eq. 16.1 uses a log-log form that produces the mildest gradient of the three.

4.3 Constraints on Interpretation

1. **Stress concentration via Y_{Sa} only.** The analysis omits a separate K_f factor because ISO 6336 Y_{Sa} already accounts for geometric stress concentration at the tooth root. A separate fatigue notch factor (*not* a geometric stress raiser) could be added to model material notch sensitivity, but must be applied to a nominal stress computed *without* Y_{Sa} to avoid double-counting.

2. **Run-out exclusion.** The 28 run-out entries (19.4% of combinations) are excluded from the ANOVA because $\log_{10}(\infty)$ is undefined. A survival-analysis framework would capture both regimes without discarding data.

3. **Quenched-and-tempered steel only.** Different materials will produce different factor rankings.

4. **No residual stress.** Case-hardened gears with compressive residual stresses would experience a lower effective stress ratio, likely reducing the mean-stress contribution.

5. **Single gear geometry and load.** Results apply to one spur gear pair at one torque level.

6. **No experimental validation.** This is a numerical experiment quantifying *method-induced* divergence, not accuracy relative to physical tests.

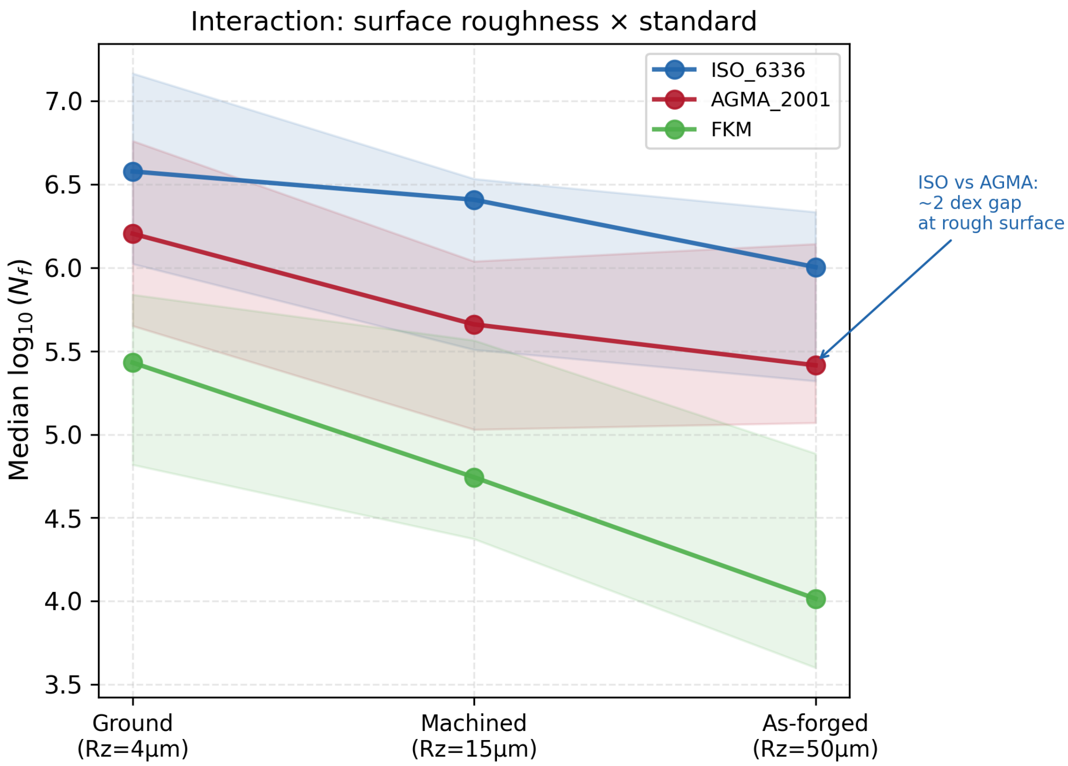

Figure 3: Interaction of surface roughness (R_z) and standard. Median $\log_{10}(N_f)$ by standard at each R_z level. Shaded bands show interquartile range.

5 Conclusions

We applied a full-factorial ANOVA to decompose the total log-variance in gear tooth bending fatigue life prediction into five methodological sources. For a quenched-and-tempered 42CrMo4 spur gear under constant-amplitude pulsating bending ($R = 0$), across 144 combinations, we find:

1. The choice of standard (ISO 6336, AGMA 2001, FKM) and the choice of mean-stress correction (Goodman, Gerber, Morrow, SWT) each account for approximately 31% of total log-variance and are statistically indistinguishable. Surface roughness (8.5%) is a clear third factor.

2. The S–N curve source (0.7%) and cumulative damage rule (0.5%) are negligible in this regime—the two Waterloo datasets, despite different Basquin parameters, produce similar relative rankings of the other methodological choices.

3. Predicted fatigue life spans from 3.3×10^2 to 3.1×10^7 cycles (5.0 dex). The same gear, run through defensible engineering workflows, can be predicted to fail at 330 cycles or survive 30 million—depending on which handbook the engineer opens and which correction formula they apply.

4. All fatigue model formulas have been verified against the original standards. The open-source pipeline is released as a reusable, auditable toolkit.

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